A SPLIT MORPHISM SPLITS ITS SOURCE INTO THE KERNEL OF THE PROJECTION AND THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the only computation it needs is stated on the first line and the two identifications it uses are named, so no step is left to the reader.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Splitting of an idempotent endomorphism. Same chapter, Lemma lemma-idempotent-splitting. Supplies the direct sum decomposition of an object on which an idempotent and its complement act, which is applied to the composite of the two given morphisms.
- The kernel and cokernel of a split endomorphism. Same chapter, Lemma lemma-split-morphism-kernel-cokernel. Supplies the identification of the first given morphism as the kernel and of the second as the cokernel of the endomorphism, which is what turns the abstract decomposition into the displayed one.

Source reference. Original lemma 0H9Q: sources/stacks-homology/pinned-original-source.tex, source lines 639–639. Navigation pointer invoked at proof line 717; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0H9R: sources/stacks-homology/pinned-original-source.tex, source lines 667–667. Navigation pointer invoked at proof line 718; the counted span contains the pointer only and the cited result is not reproduced

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inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be a preadditive category, x and y be objects of $\mathcal{A}$, and $j : y \to x$ and $q : x \to y$ be morphisms satisfying $q \circ j = id_y$. If $j \circ q$ has at least one of a kernel or cokernel, then both (co)kernels exist and*

$$\begin{aligned} x &\simeq \mathrm{Ker}(j \circ q) \oplus y \\ &\simeq \mathrm{Coker}(j \circ q) \oplus y \end{aligned}$$

with the structure of either direct product including j, q, and the canonical (co)kernel morphism, and the remaining map uniquely determined.

Proof. Write $f = j \circ q$, compute that $f \circ f = f$, and apply Lemma 0H9Q and Lemma 0H9R. □